



GBC1, WS 2025

Chlorophyll Determination

Determination of Chlorophyll Concentration in Biomatter using Spectrophotometry

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1 Background

1.1 Chlorophyll

Chlorophyll is a green pigment found in plants and some bacteria. It is responsible for the green color of plants and is essential for photosynthesis. It consists of two main types: Chlorophyll a (Chl_a) and Chlorophyll b (Chl_b).

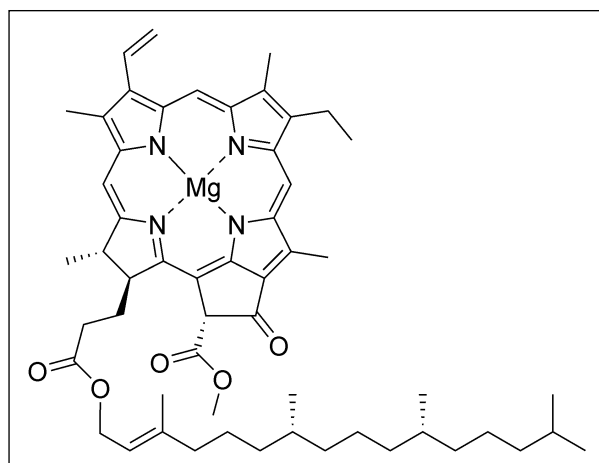


Figure 1: Chlorophyll A structure ^[1]

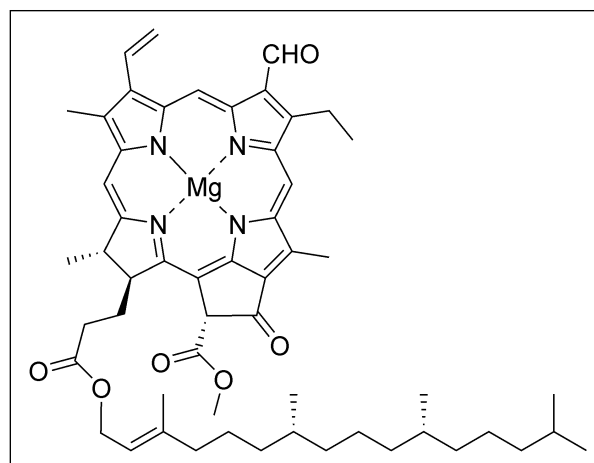


Figure 2: Chlorophyll B structure ^[2]

It is the foundation of life for plants, algae, and cyanobacteria, by converting sunlight into chemical energy. It is a key component of the photosynthetic process and is essential for the production of oxygen and food. ^[3]

1.1.1 Solubility

Chlorophyll has a long fatty chain called phytol ($\text{C}_{20}\text{H}_{39}$) attached to it. This chain makes chlorophyll dissolve well in oils and organic solvents, but not in water. The phytol chain acts like an anchor that holds the chlorophyll in place within the plant's membrane structures.

Because of this fatty chain, organic solvents like acetone, methanol, or ethanol are needed to extract chlorophyll from plant tissues. 80% acetone (mixed with 20% water) is commonly used because the small amount of water helps break apart the protein structures that hold the chlorophyll. ^{[4], [5]} 100% acetone is very suitable for leaf extracts. ^[4]

1.1.2 Solvatochromism

The solvent effect, known as "Solvatochromism", is a phenomenon that occurs when the absorption spectrum of a compound changes between different solvents used. This solvent-dependent effect applies to chlorophyll and arises from the interaction between the solvent molecules and the chlorophyll's π -electron cloud ^{[3], [6]}.

Consequently, absorption coefficients derived for methanol cannot be used for acetone extracts, and coefficients for 100% acetone cannot be used for 80% acetone. The specific equations discussed in Section 3.2 are strictly valid only for 100% Acetone extracts. ^{[3], [4]}

1.1.3 Photophysics and structural distinctions

Chlorophylls are Mg-coordinated chlorins (porphyrin derivatives) with an extended conjugated π -system that gives rise to two dominant absorption regions: an intense Soret band in the blue

(~ 430 – 460nm) and a weaker Q band in the red (~ 640 – 670nm) region. The weak absorption in the green explains the perceived color of foliage. ^[3]

The only structural difference between Chl_a and Chl_b at C – 7 (methyl vs. formyl) shifts the spectra: Chl_a peaks near 664nm and Chl_b near 647nm, enabling simultaneous quantification in mixtures ^{[1], [2]}. The hydrophobic phytol tail anchors chlorophylls in thylakoid membranes and dictates solubility in organic solvents ^[5].

1.2 Spectrophotometry

1.2.1 Principle

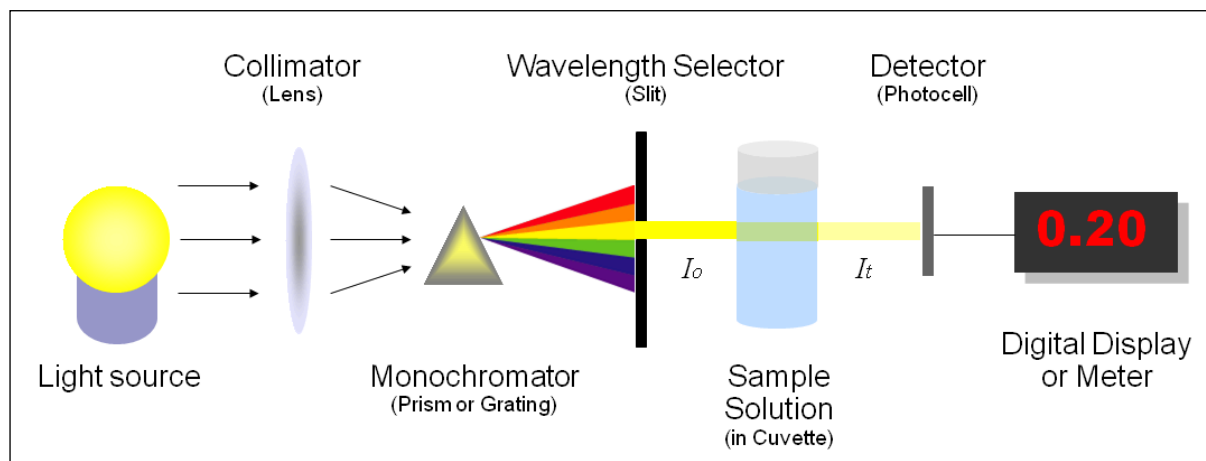


Figure 3: Principle of spectrophotometry (CC BY-4.0; Heesung Shim via LibreTexts) ^[7]

Spectrophotometry is an analytical technique used to quantify the concentration of a substance by measuring its ability to absorb light at specific wavelengths. When a beam of monochromatic light with an initial intensity I_0 passes through a sample, the molecules within the solution interact with the photons, leading to a reduction in intensity to a transmitted value I . This interaction is governed by the electronic structure of the analyte, where photons are absorbed if their energy matches the electronic transition states of the molecule, as discussed in Section 1.1.2. ^[7]

The relationship between the incident and transmitted light is expressed through transmittance T and absorbance A :

$$(1) \quad T = \frac{I}{I_0}$$

$$(2) \quad A = -\log_{10}(T)$$

While transmittance describes the fraction of light passing through the sample, absorbance is the preferred metric in quantitative analysis because of its linear relationship with concentration c and path length d , as stated by the Beer-Lambert law. This linearity typically holds within a specific concentration range, beyond which chemical or instrumental deviations occur. In standard laboratory settings, cuvettes with a path length of 1cm are utilized to simplify the mathematical conversion of absorbance into concentration ^[7].

1.2.2 General process

The experimental procedure begins with selecting the diagnostic wavelengths relevant to the pigments of interest—most notably the red absorption maxima for chlorophylls—and allowing the spectrophotometer to warm up to ensure a stable light source. A baseline is established by “blanking” the instrument using a cuvette filled with pure solvent (e.g., 80% acetone), which defines I_0 and accounts for any absorption or reflection by the cuvette and solvent itself ^[7].

Once the baseline is set, the sample extract is measured to determine the transmitted intensity I , from which the absorbance A is calculated. It is critical to ensure that the measured absorbance stays within the instrument's linear dynamic range (typically between 0.1 and 1.0); samples exceeding this range must be diluted and remeasured to preserve accuracy. For mixtures like leaf extracts, measurements are taken at multiple wavelengths (e.g., 647nm and 664nm) to resolve the individual contributions of Chl_a and Chl_b using simultaneous equations based on the principle of additivity (see Equation 4). In cases where turbidity or light scattering from cell debris is suspected, a measurement at a non-absorbing wavelength such as 750nm is performed to subtract the background noise from the pigment readings. ^{[4], [5]} Finally, the calculated concentrations in the extract are converted to fresh weight values by accounting for the initial sample weight and extraction volume ^[7].

1.2.3 Beer-Lambert law

The Beer-Lambert law, shown in Equation 3 and based on Equation 1, describes the correlation between the dimensionless total absorbance E_λ , the molar extinction coefficient ϵ_λ ($\frac{\text{l}}{\text{mol} \cdot \text{cm}}$ ^[8]) and concentration c ($\frac{\text{mol}}{\text{l}}$) of the substance of interest in the sample, and the light path length d (cm) through the sample, where λ is an arbitrary wavelength of light.

$$(3) \quad E_\lambda = \epsilon_\lambda \cdot c \cdot d$$

For example Chl_a has $\epsilon_{420} = 85\,000 \text{ l mol}^{-1} \text{ cm}^{-1}$. This, however, depends on a multitude of factors such as pH, temperature, or solvent, and is therefore not a constant ^[8].

1.2.4 Mixtures and additivity

In an extract, total absorbance at a given wavelength is the sum of each pigment's absorbance:

$$(4) \quad E_\lambda = d \cdot \sum_i \epsilon_{i,\lambda} \cdot c_i$$

Equation 4 must be taken into account when calculating concentrations of real world samples, since there is rarely any purely isolated sample available. ^[9]

1.2.5 Calculation of Chlorophyll Concentration

By rearranging Equation 3 we get:

$$(5) \quad c \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_\lambda}{\epsilon_\lambda} \cdot \frac{1}{d}$$

d is the path length through the sample or in other words the thickness of the cuvette. The global standard for this is $d = 1 \text{ cm}$ ^[10]. Therefore, we can effectively cancel out d in the equation.

While Equation 5 is a general equation yielding $\frac{\text{mol}}{\text{l}}$, we can convert to mass concentration $\frac{\text{g}}{\text{l}}$ by multiplying with the molar mass MG_S of our substance S and factoring in the dilution factor $\frac{V_E}{V_P}$, where V_E is the volume of the extraction solvent and V_P is the volume of the biological source material.

$$(6) \quad c \left[\frac{\text{g}}{\text{l}_E} \right] = \frac{E_\lambda}{\epsilon_\lambda} \cdot MG_S \cdot \frac{1}{d} \Rightarrow$$

$$c \left[\frac{\text{g}}{\text{l}_P} \right] = c \left[\frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{V_P}$$

Here $c \left[\frac{\text{g}}{\text{l}_E} \right]$ is the mass concentration of the substance S in the extraction solvent and $c \left[\frac{\text{g}}{\text{l}_P} \right]$ is the mass concentration of the substance S per volume of the biological source material.

Note: We can use the same approach as for removing the dilution factor $\frac{V_E}{V_P}$ to convert from $\frac{\text{g}}{\text{l}_E}$ to $\frac{\text{mg}}{\text{g}_P}$ by multiplying with the dilution factor $\frac{V_E}{M_P}$, as shown in Equation 7.

$$(7) \quad c \left[\frac{\text{mg}}{\text{g}_P} \right] = c \left[\frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{M_P} \cdot 1000$$

Where M_P is the mass of the biological source material in g, and $c \left[\frac{\text{mg}}{\text{g}_P} \right]$ is the mass of the substance S per weight of biological source material. This is often used to determine the concentration in fresh weight.

2 Execution

The experiment was performed in two variants, where the first variant was performed according to the instructor's guidance. In this variant, a diluted chlorophyll solution was used as reference. The second variant will be described going forward. Figure 4 displays the experimental workflow for this second variant.

2.1 Preparation

2.1.1 Materials needed

According to the instruction sheet, we need the following materials:

- Scissors
- Mortar and pestle
- Spatula
- Paper filter
- Funnel
- Beaker glass
- Glass cuvettes
- Spectrophotometer

Chemicals:

- Acetone (~ 20ml)
- Sea sand
- Calcium carbonate
- Plant material (2 – 5g)

Instead of sea sand and calcium carbonate, we used quartz sand for the mortar, as the experiment was set up that way in our lab session.

2.1.2 Chlorophyll extraction

About 2 – 5g of fresh plant material was cut up with scissors and put into a mortar. A small amount (tip of a spatula) of sea sand (or quartz sand, as noted in Section 2.1.1) and calcium carbonate (CaCO_3) was added to help break down the plant cells and neutralize acids. The plant material was then ground thoroughly with a pestle to break open all the cells.

Next, about 20ml of acetone was added, and the mixture was ground further until a dark green liquid formed. This chlorophyll extract was then filtered through filter paper into a beaker. The filter paper was folded twice to make a cone shape that fit into the funnel.

2.2 Spectrophotometric measurement

The chlorophyll concentration was measured using a spectrophotometer. First, the machine was calibrated (Base Corr) using a glass cuvette filled with pure acetone to set the baseline from 400 – 800nm. This step removes the effect of the acetone and cuvette on the measurements. After calibration, the chlorophyll sample was measured across the same wavelength range to get the full absorption spectrum.

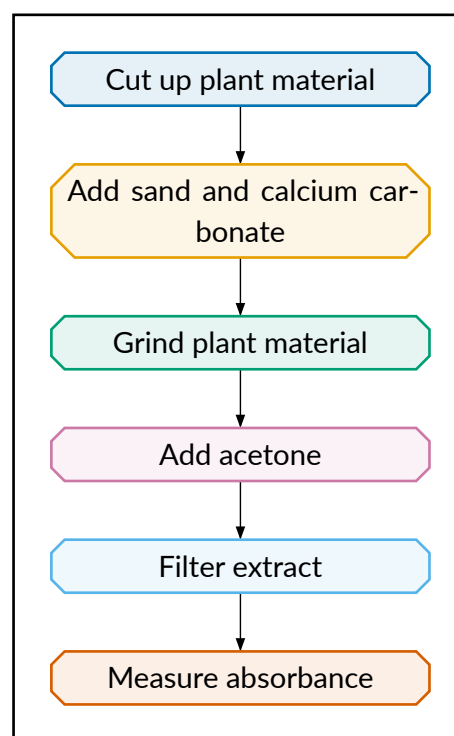


Figure 4: Experimental workflow for the extraction and isolation of crude chlorophyll from plant tissue.

2.3 Explanation

The specific methodology employed in this experiment serves to maximize extraction efficiency and ensure the chemical stability of the pigments:

- **Mechanical Disruption:** The use of scissors for comminution and sand during mortar homogenization provides the necessary abrasive force to breach the rigid cellulosic cell walls of the plant material. This increases the surface area for the solvent to interact with the intracellular components.
- **Neutralization with CaCO_3 :** When plant cells are crushed, organic acids are released from the vacuoles. An acidic environment promotes the loss of the central magnesium ion (Mg^{2+}) from the chlorophyll porphyrin ring, converting it into pheophytin. This conversion significantly alters the absorption spectrum, leading to inaccurate quantification. The addition of calcium carbonate neutralizes these acids, preserving the chlorophyll in its native state.
- **Cuvette Material:** Standard plastic cuvettes (typically made of polystyrene or PMMA) are chemically incompatible with acetone, which acts as a solvent that can dissolve or cloud the plastic. Using glass cuvettes is mandatory to maintain optical clarity and structural integrity throughout the measurement.
- **Base Correction:** Calibration with pure acetone accounts for any light absorption or reflection caused by the solvent itself and the glass-air interfaces of the cuvette. By subtracting this baseline, the resulting spectrum represents the absorbance of the dissolved pigments alone.

3 Results

Various analysis tasks were performed to better understand the data and the results. These were done in isolation to avoid any confounding factors.

We decided to take a detailed full spectrum sample using a mixture of brussels sprouts and maggi herbs ¹. We will use this sample for all single sample analysis, especially in Section 3.2 and Section 3.3. We refer to this sample as “ours” going forward.

3.1 Spectral Analysis

The full spectrum data was derived by sampling absorption rates at 10nm intervals from the spectrophotometer's automatic full spectrum curve. The results are shown in Figure 6.

3.1.1 Reference spectrum

We compare our full spectrum data with the reference spectra of chlorophyll a and chlorophyll b in diethyl ether. This is used for reference even though we used acetone in our experiment, because we failed to find other publicly available full spectrum reference data.

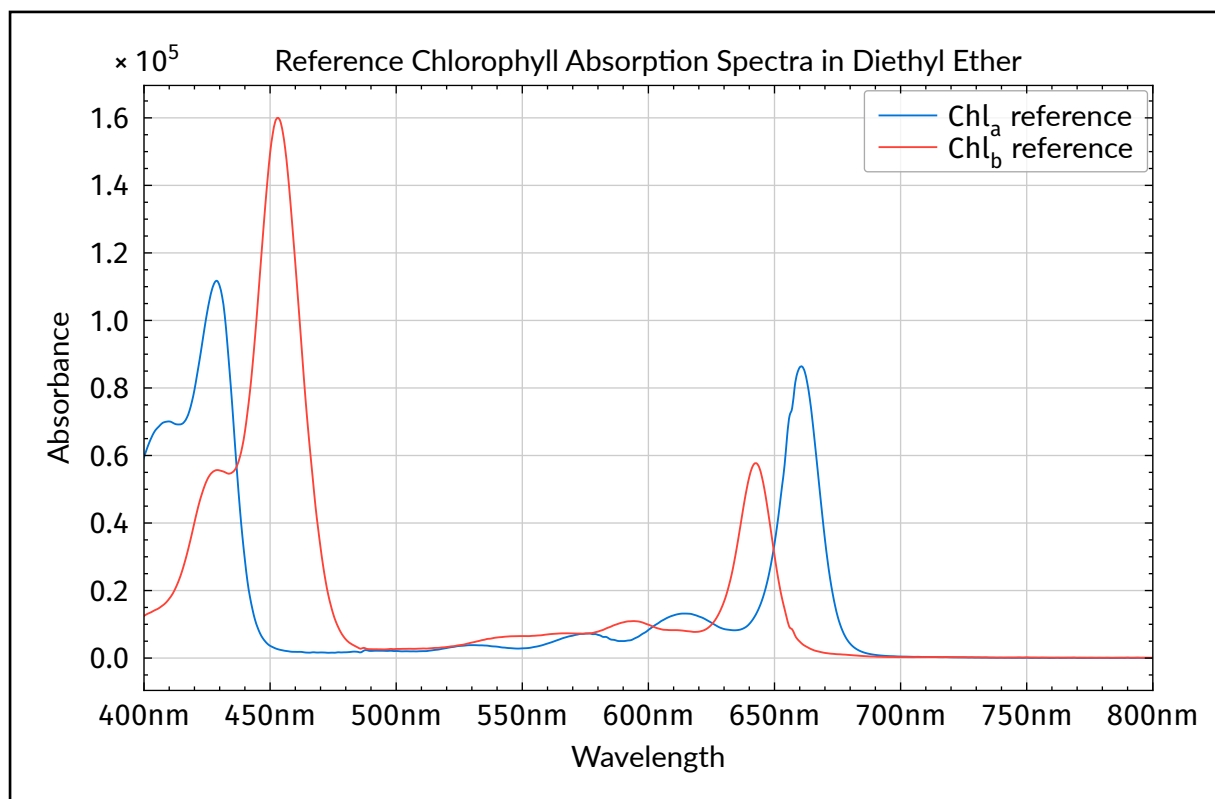


Figure 5: Reference Chlorophyll Absorption Spectra in Diethyl Ether ^{[1], [2]}. Blue shows Chl_a and red shows Chl_b

In Figure 5 we can observe the blue-region spectrum peaks at ~ 430nm for Chl_a and ~ 452nm for Chl_b. The red-region spectrum peaks at ~ 660nm for Chl_a and ~ 642nm for Chl_b.

These match the characteristics of the chlorophyll absorption spectra in diethyl ether. We can observe that the spectrum of Chl_a shows a characteristic shoulder when crossing the line of Chl_b in the blue region, which is also characteristic for both chlorophyll types. ^[3]

¹File group 1, subgroup 6 from the results sheet

Therefore, we conclude the found dataset to be a viable and accurate source of reference.

3.1.2 Sample spectrum

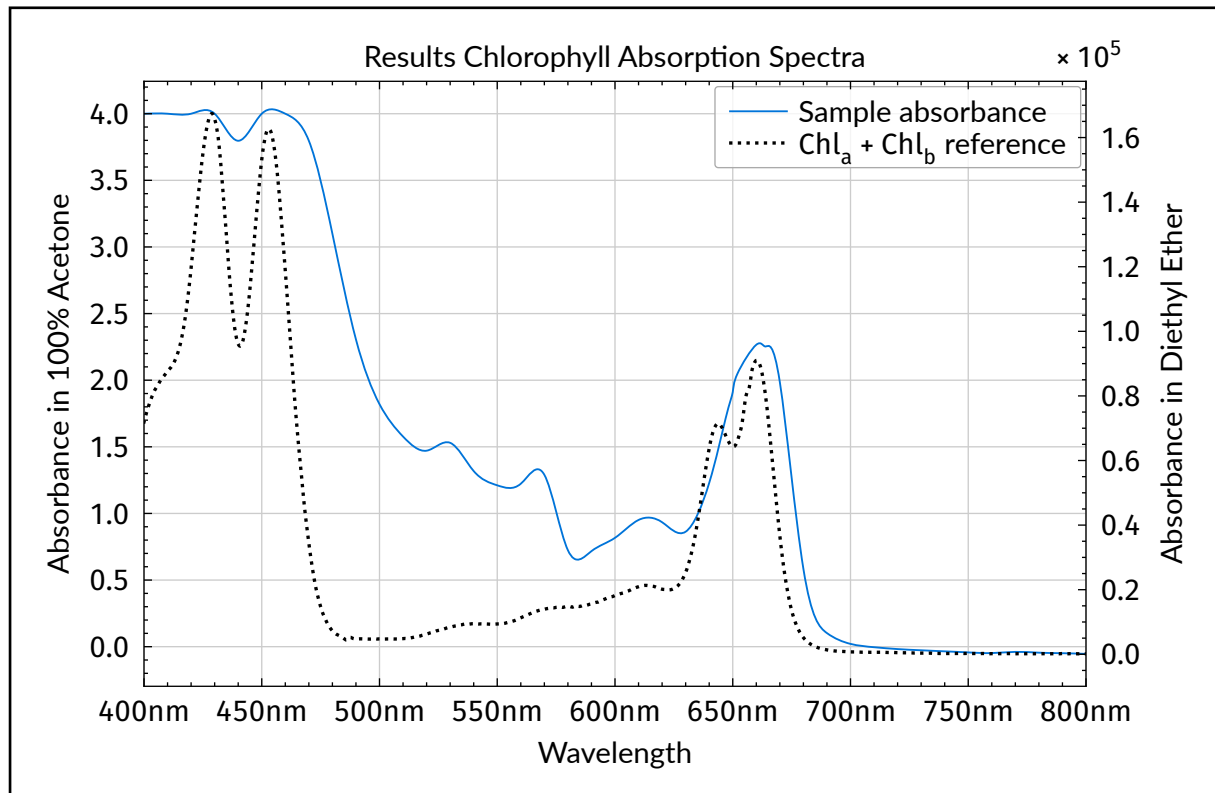


Figure 6: Absorption Spectrum of the Sample (shown in blue) and combined $\text{Chl}_a + \text{Chl}_b$ Reference from Figure 5 (shown in dotted black)

The sample spectrum shows the absorption of the total, non-isolated chlorophylls ($\text{Chl}_a + \text{Chl}_b$) in the sample. We can observe three peaks: two in the blue region at $\sim 430\text{nm}$ and $\sim 445\text{nm}$ with one in the red region at $\sim 660\text{nm}$.

We can observe similar peaks as in the reference spectrum, but with slight distortions and more or less pronounced features. This is because the sample consists of total, non-isolated chlorophylls ($\text{Chl}_a + \text{Chl}_b$) and possibly other pigments dissolved in acetone. Furthermore, as mentioned in Section 1.1.2, shifts are to be expected when comparing our sample with diethyl ether dissolved reference^[3]. Despite this, one can even spot fine features in our spectrum. For example, the valley at the $\sim 650\text{nm}$ wavelength that is clearly visible in the reference spectrum is present in our data in the form of a small bump.

The most glaring difference compared to reference is the sample's gradually increasing absorption in the green spectrum (read from right/red to left/blue), whereas according to the reference the absorption should slightly decrease. One possible explanation for this is the presence of other pigments in the sample that absorb in the green spectrum, such as carotenoids^{[3], [5]}. Another plausible explanation is incomplete calibration: Calibration did not happen across the entire spectrum but only once on the higher end of the wavelength. The level of unfamiliarity with the spectrophotometer might have contributed to this.

3.2 Chlorophyll Concentration with Instructor's Formula

The instructor provided us with a formula to calculate the chlorophyll concentration:

$$\begin{aligned}
 (8) \quad c_a \left[\frac{\text{mg}}{\text{l}} \right] &= 11.78 E_{664} - 2.29 E_{647} \\
 c_b \left[\frac{\text{mg}}{\text{l}} \right] &= 20.05 E_{647} - 4.77 E_{664} \\
 c_{\text{total}} &= c_a + c_b = 27.8 E_{652}
 \end{aligned}$$

Plugging in the concrete values, shown in Table 1, for our sample we get:

$$\begin{aligned}
 c_a \left[\frac{\text{mg}}{\text{l}} \right] &= 11.78 \cdot 2.253 - 2.29 \cdot 1.715 = \mathbf{22.61299} \\
 c_b \left[\frac{\text{mg}}{\text{l}} \right] &= 20.05 \cdot 1.715 - 4.77 \cdot 2.253 = \mathbf{23.63894} \\
 c_{\text{total}} \left[\frac{\text{mg}}{\text{l}} \right] &= c_a + c_b = 22.61299 + 23.63894 = \mathbf{46.25193} \\
 &= 27.8 E_{652} = 27.8 \cdot 2.049 = \mathbf{56.9622}
 \end{aligned}$$

Wavelength	Absorbance
664	2.253
647	1.715
652	2.049

Table 1: Wavelengths and absorbance of the results from our sample needed for calculating the chlorophyll concentration using the given formulas

A quick sanity check with the Gemini LLM for the plausibility of these values was positive. Lab extracts have a common range of $10 - 30 \frac{\text{mg}}{\text{l}}$. This aligns with our values.

We can see a significant difference between the total result using the standalone absorbance factor and the addition of Chl_a and Chl_b , with an absolute difference of $10.7103 \frac{\text{mg}}{\text{l}}$ and a relative difference of 18.8 %.

3.2.1 Interpretation of the difference

It proves difficult to find a possible explanation for this discrepancy without knowing where the given formulas originate from and how they were derived. Measurement or execution errors are a possibility, although this seems unlikely due to the high difference.

3.3 Chlorophyll Concentration using Beer-Lambert Law

In order to find both the Chl_a and Chl_b concentration, we need to use the Beer-Lambert law. We arrange Equation 3 and Equation 4 into a linear equation system to solve for both concentrations:

$$\begin{aligned}
 (9) \quad E_{664} &= \varepsilon_{a,664} \cdot c_a + \varepsilon_{b,664} \cdot c_b \\
 E_{647} &= \varepsilon_{a,647} \cdot c_a + \varepsilon_{b,647} \cdot c_b
 \end{aligned}$$

Because we use the standard path length of $d = 1\text{cm}$ we can ignore $\frac{1}{d}$ in our calculations. Using substitution we can solve for c_a and c_b :

$$(10) \quad c_a \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_{664} \epsilon_{b,647} - E_{647} \epsilon_{b,664}}{\epsilon_{a,664} \epsilon_{b,647} - \epsilon_{a,647} \epsilon_{b,664}}$$

$$c_b \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_{647} \epsilon_{a,664} - E_{664} \epsilon_{a,647}}{\epsilon_{a,664} \epsilon_{b,647} - \epsilon_{b,664} \epsilon_{a,647}}$$

We arrive at the final formula for calculating the total chlorophyll concentration, accounting for the mass concentration and dilution factor:

$$(11) \quad c_a \left[\frac{\text{g}}{\text{l}_E} \right] = c_a \left[\frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_a}$$

$$c_b \left[\frac{\text{g}}{\text{l}_E} \right] = c_b \left[\frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_b}$$

$$c_{\text{total}} \left[\frac{\text{g}}{\text{l}_E} \right] = c_a \left[\frac{\text{g}}{\text{l}_E} \right] + c_b \left[\frac{\text{g}}{\text{l}_E} \right]$$

$$c_{\text{total}} \left[\frac{\text{g}}{\text{g}_P} \right] = c_{\text{total}} \left[\frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{M_P}$$

By plugging the values from Table 2 into these equations, we arrive at the final values for the mass concentration in the extraction solvent:

$$c_a \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{23.38}$$

$$c_b \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{24.27}$$

$$c_{\text{total}} \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{47.65}$$

Accounting for the dilution factor, the final determined mass concentration in the biological source material (fresh weight) is:

$$c_{\text{total}} \left[\frac{\text{mg}}{\text{g}_P} \right] = c_{\text{total}} \left[\frac{\text{mg}}{\text{l}_E} \right] \cdot \frac{V_E}{M_P} = \mathbf{0.4984}$$

3.3.1 Comparison to instructor's formula

The values calculated using the Beer-Lambert law are in line with the values calculated using the instructor's formulas in Section 3.2. The difference is small (2.93 %) and indicates that the Beer-Lambert law and our calculations are a valid method for determining the chlorophyll concentration. Small differences can be attributed to different extinction coefficients of the chlorophylls used when precalculating the given formulas' factors.

	MG_S	ϵ_{647}	ϵ_{664}	E_{647}	E_{664}	$V_E \div M_P$
Chl_a	893.5 g mol ⁻¹	17 464 l mol ⁻¹ cm ⁻¹	76 790 l mol ⁻¹ cm ⁻¹	1.715	2.253	$\frac{19.98\text{ml}}{1910\text{mg}} =$ 0.01046
Chl_b	907.5 g mol ⁻¹	47 040 l mol ⁻¹ cm ⁻¹	9 121 l mol ⁻¹ cm ⁻¹			

Table 2: Extinction coefficients ^[11], molecular weights, absorbance values and dilution factor needed for calculating the chlorophyll concentration using the Beer-Lambert law.

Section 3.2.1 describes the unknown origin of the instructor-provided formulas. After performing the calculations ourselves, we can observe that once the extinction coefficients ϵ and molecular weights MG are known for both chlorophyll types, the concentration equation reduces to simple multilinear functions with constant factors applied to the remaining variables E_λ . This suggests that the given formulas are simplified versions of the complete calculation chain we performed based on the Beer-Lambert law.

Interestingly enough, even though the extinction coefficients ϵ used in our manual calculations are for 80% acetone solutions^[11], the Beer-Lambert law calculations still yield results in line with the instructor's formulas, despite the solvatochromism effect stated in Section 1.1.2. The small difference between the results of the instructor's formula and the Beer-Lambert law calculations could then be explained by this and the hypothesis that the absorption difference between 80% acetone and 100% acetone is not significant enough to cause a noticeable difference in the calculated concentration.

3.3.2 Checking for alignment with known concentrations

With the calculated concentration value in fresh weight for our sample, we want to check for plausibility by comparing to other well known concentrations of common leafy vegetables in fresh weight.

Chlorophyll content varies widely by species, leaf age, and sunlight exposure.

- Total Chlorophyll (a+b): Generally ranges from 0.08 to 19.2 mg/g fresh weight.
- Common Leafy Vegetables:
 - Spinach: 120-150 mg/100g (1.2-1.5 mg/g).
 - Chard: 121 mg/100g.
 - Lettuce: 19-58 mg/100g.
 - Parsley: 5-20 µg/g.

— Gemini LLM 2025

Chlorophyll Content Comparison:

Lovage generally has higher chlorophyll concentrations than most lettuce varieties but slightly lower levels than the most dense leafy herbs like parsley.

- **Lovage (Maggi Herb):** 0.68 mg/g (68.5 mg/100g)
- **Brussels Sprouts:** 2.41 mg/g
- **Lettuce:** 0.12 - 1.0 mg/g
- **Parsley:** 2.18 - 2.23 mg/g

— Gemini LLM 2025

Using this information and normalizing to mg/g of fresh weight, the results of which are shown in Table 3, we can compare our results in fresh weight to the known concentrations of common leafy vegetables.

Spinach	Chard	Lovage (Maggi Herb)	Brussels Sprouts	Lettuce	Parsley
1.2 - 1.5	1.21	0.68	2.41	0.12 - 1	2.18 - 2.23

Table 3: Known concentrations of common leafy vegetables, queried from Gemini 2025, normalized to mg/g of fresh weight.

Note: Even though Gemini is no scientific source, we only use it for a basic plausibility check of our own results. It can therefore be deemed a reliable enough source for this specific usecase.

The total chlorophyll concentration of our sample is 0.4984 mg/g of fresh weight. While lower than some literature values, it is within the expected physiological range and thus deemed plausible. We do not suspect a great systematic error in our measurements, but rather a combination of multiple smaller errors. These could include inadequate processing/grinding with quartz sand, the fact that the material was cooled and partially frozen during preparation, incomplete extraction due to insufficient contact time with the solvent, or degradation of chlorophyll during sample handling.

3.4 Descriptive Statistics

We analyze the results of all groups in our class and compare them to our own results. We also compare different source materials with each other. Unless explicitly stated otherwise, the concentrations in fresh weight are computed using the instructor's formula for individual chlorophyll types and adding them together, after which dilution compensation is performed. We do not use the pre-calculated concentrations from the results sheet.

3.4.1 Overall results

Mean $\bar{x}$	Standard Deviation $\text{std}(x)$	Median $\tilde{x}$	Range $\text{max}(x) - \text{min}(x)$
0.2761 mg/g	0.1978 mg/g	0.1785 mg/g	0.6896 - 0.0646 mg/g

Table 4: Descriptive statistics of the total chlorophyll concentration for all groups

Table 4 shows all descriptive statistics of the total chlorophyll concentration in fresh weight for all groups. We can use this information to put our group's results in context.

Based on the instructor's formula for the total chlorophyll concentration, our total chlorophyll concentration in fresh weight is 0.4838 mg/g. This value lies at the higher end of our data range but is not an outlier; it indicates a result consistent with our experiment. The high value makes sense, as Table 3 shows that brussels sprouts have a very high concentration of chlorophyll.

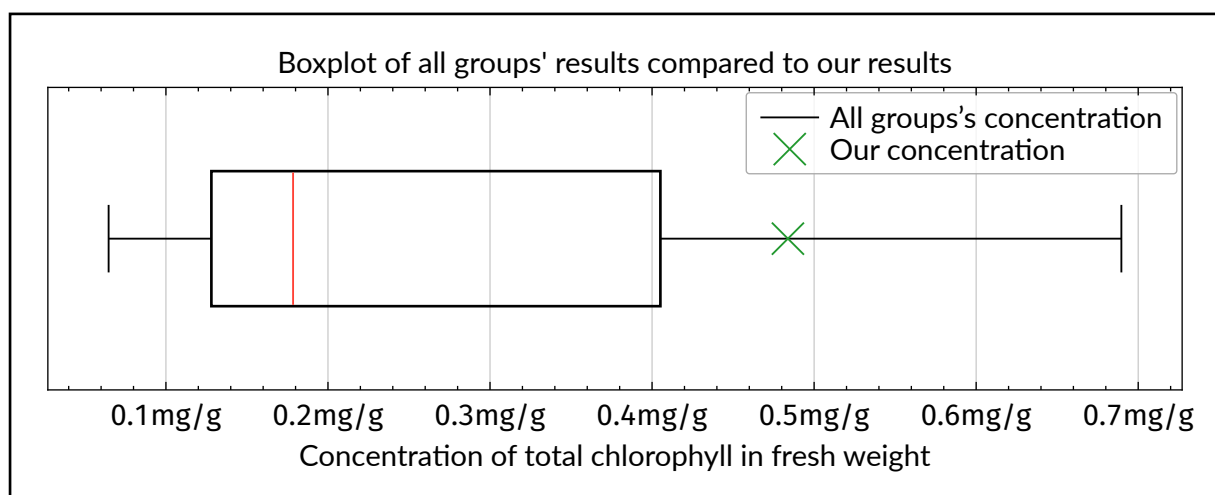


Figure 7: Boxplot of all groups' results compared to our results

Figure 7 shows a boxplot of all groups' results compared to our results. It demonstrates that our value lies between the third and fourth quartile of the data, and supports our claims above.

3.4.2 Results per source material

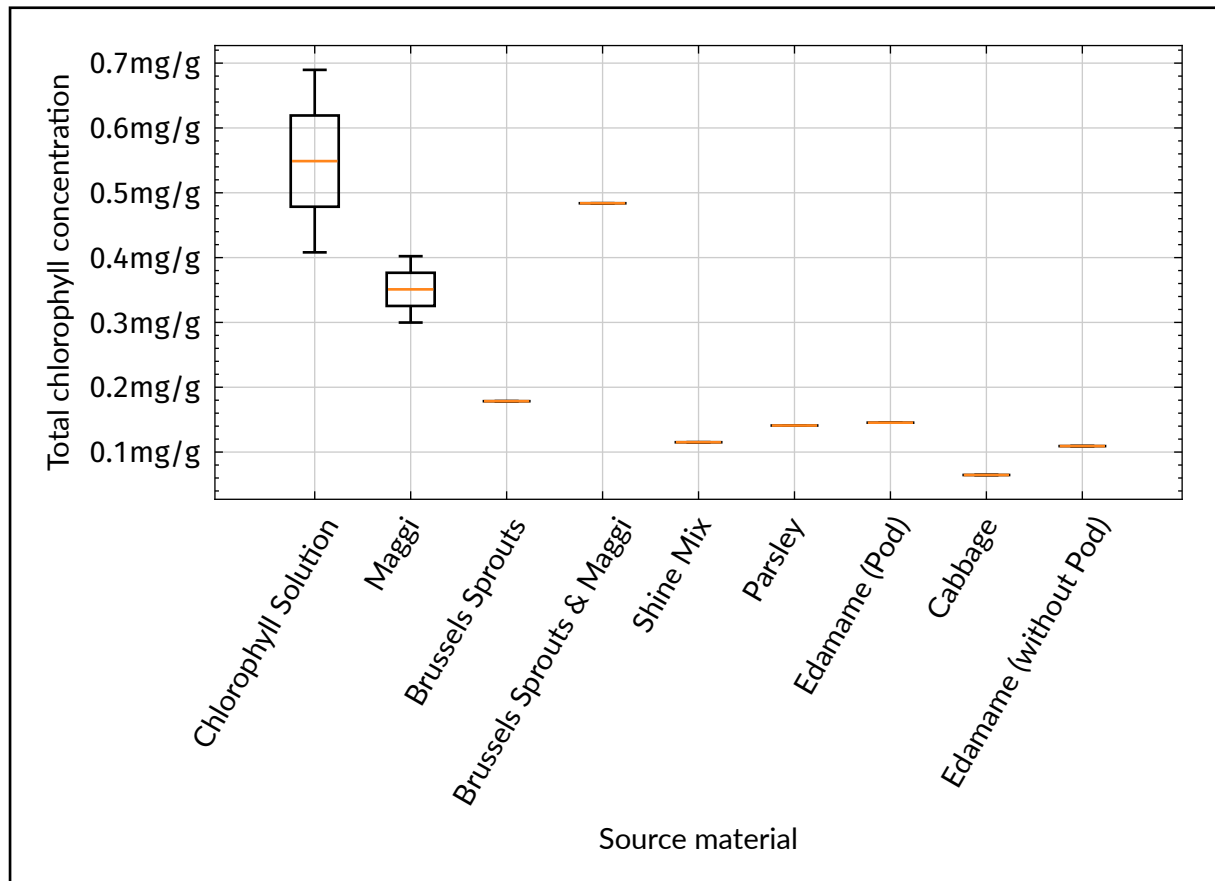


Figure 8: Boxplot of all groups' results per source material

Figure 8 shows interesting insights. Most of the source materials have only a singular entry and are therefore shown flat in the plot.

For the shine mix, parsley, edamame (with or without pods), and cabbage, the results are very similar. This is quite surprising, as Table 3 lists parsley with about $15 \times$ higher concentration than we see in the boxplot.

The brussels sprouts and maggi mix has the highest concentration of the natural source materials (which excludes the chlorophyll solution). This should not be the case, as it should be somewhere in between the concentration for pure maggi herbs and brussels sprouts, yet it exceeds both.

One would expect the chlorophyll solution to have a very high concentration, as theoretically there should be a fresh weight concentration of 1mg/g (due to assuming 0.1% solution to be the fresh weight). We can see that trend reflected in the boxplot, even though the median is below the theoretical value.

This deviation is not unique to the chlorophyll solution, as virtually all natural source materials have a lower concentration than their expected value, based on Table 3.

3.4.2.1 Explanation for the deviation

Several factors explain why our measured values may be lower than expected:

- **Incomplete extraction:** The acetone may not have extracted all chlorophyll from the plant cells. Different plants have different cell wall structures, making some harder to extract from than others.

- **Chlorophyll breakdown:** Chlorophyll is sensitive to light and heat. Some may have broken down during extraction and measurement, reducing the final concentration.
- **Measurement errors:** Small errors in weighing samples, measuring volumes, or reading the spectrophotometer can add up and affect the results.
- **Natural variation:** Real plants vary in chlorophyll content depending on age, growing conditions, and storage. The reference values are averages that may not match our specific samples.

We do not conclude that the calculations or values are catastrophically wrong; although there are wildly different raw data points, the overall results of the calculation chain seem to settle at sensible values, as illustrated in Figure 9. Instead, we assume that the error sources are small but consistent across all groups.

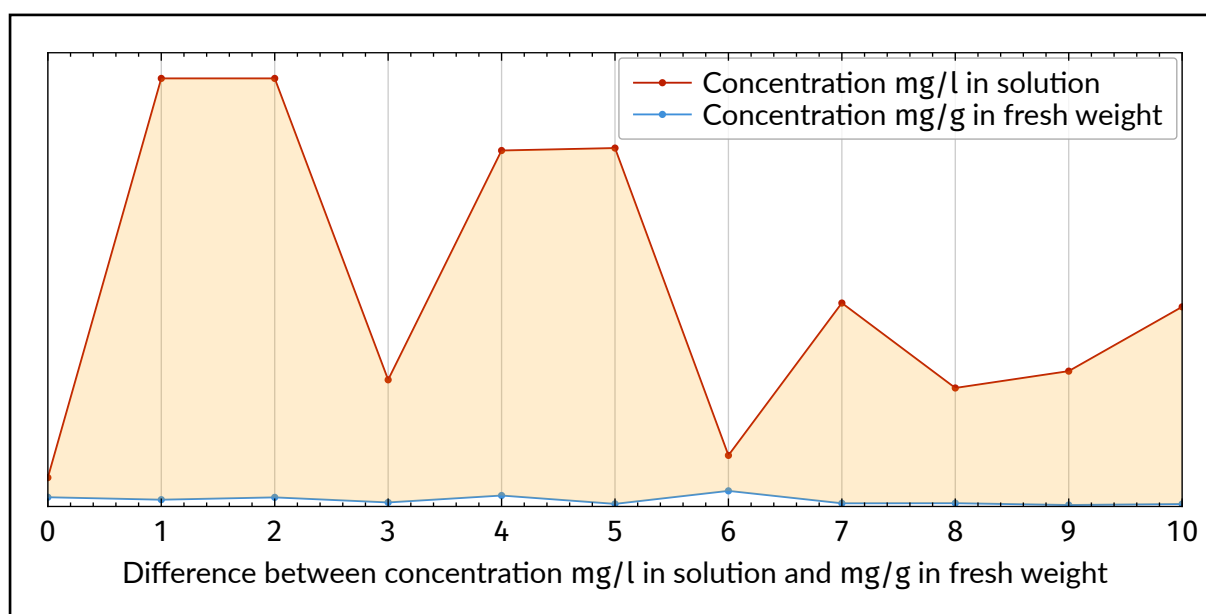


Figure 9: Difference between chlorophyll concentration in solution and in fresh weight for all groups, showing the diverge

Data Points	Standard Deviation	Normalized Standard Deviation
Concentration mg/l in solution	± 23.527 mg/l	± 0.6645
Concentration mg/g in fresh weight	± 0.1978 mg/g	± 0.7165

Table 5: Standard deviation of the data points for concentration in solution and in fresh weight

We can also compare the standard deviation of the soluted data points and the fresh weight data points. Table 5 shows that the compensation of the dilution factor works well, since we can effectively map all data points to a smaller more concrete range. This, however, is not perfect, as the *normalized*² standard deviation of the fresh weight data points is slightly higher than that of the solution data points. This shows that when accounting for unit differences, the fresh weight data points are more scattered than the solution data points.

3.4.3 Comparison between fresh weight concentration sources

As mentioned in the introduction of Section 3.4, until now we have ignored the filled out concentrations on the results sheet, and instead recalculated using the instructor's formula based on the written down extinction coefficients. We will now compare these freshly recalculated

²normalized: data points divided by their common mean

concentrations with the concentrations filled out on the results sheets. For this purpose, we take the Chl_a and Chl_b concentrations filled out on the results sheet, add them together and compensate for the dilution factor to get the total chlorophyll concentration in fresh weight.

Afterwards, we compare the freshly recalculated total concentration with a total concentration calculated using the instructor's formula based on the 652nm extinction coefficient.

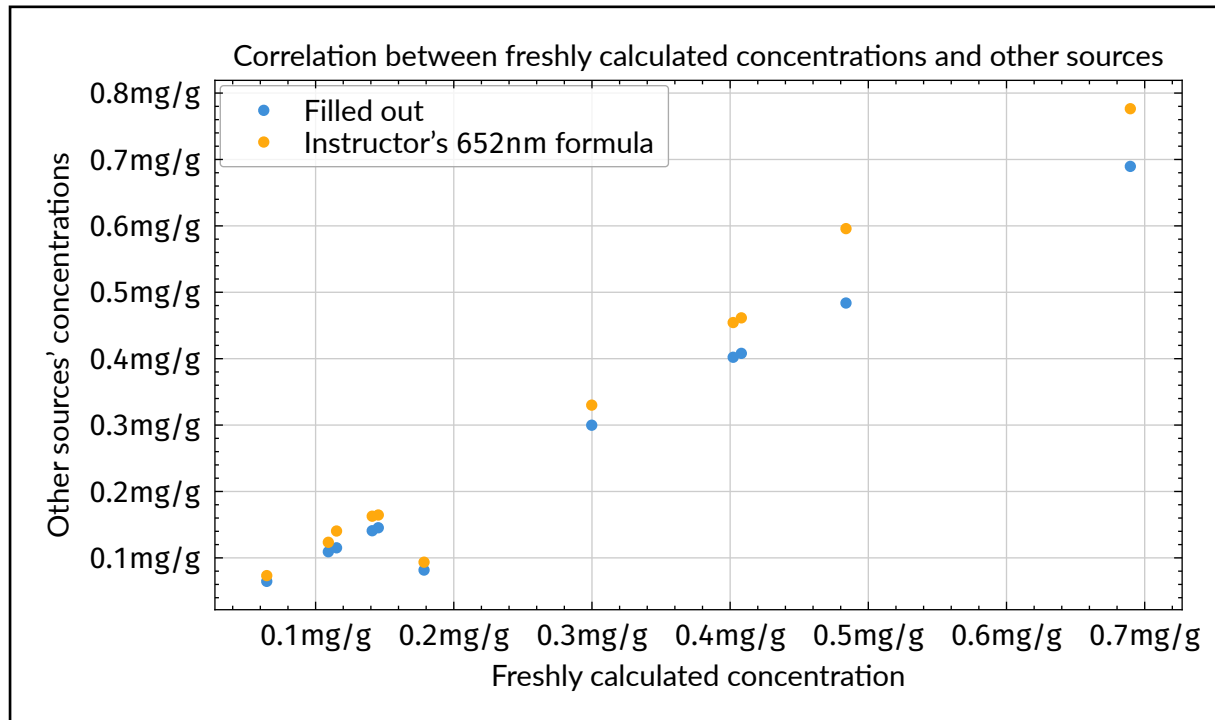


Figure 10: Comparison between freshly calculated concentrations and other sources using a scatter plot for correlation visualization, showing strong linear correlation between the sources

The different calculation paths are visualized in Figure 11. It shows the different steps of the calculation process and the different sources of data used. It especially highlights the difference between individual concentrations used for the freshly calculated concentrations and filled out concentrations paths.

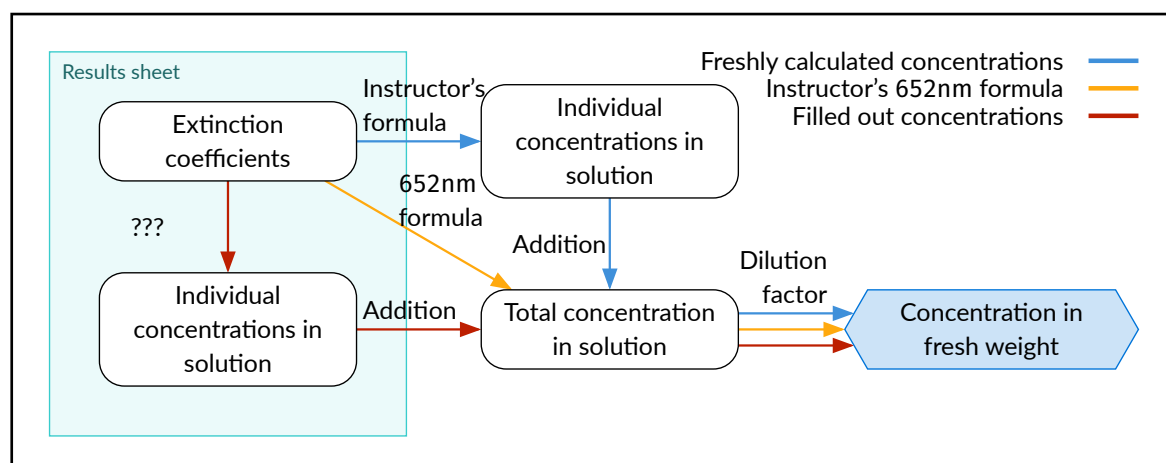


Figure 11: Diagram of the different calculation paths for the total chlorophyll concentration in fresh weight

3.4.3.1 Filled out & freshly calculated

There is almost no difference between the filled out and freshly calculated concentrations. Figure 12 shows that there is only one entry with a difference of $\sim 1\text{mg/g}$ between the two values. This discrepancy is likely due to an error in writing down the concentration values on the results sheet, rather than a systematic calculation error, as the difference is isolated to a single data point and not significant enough to indicate a fundamental issue with the calculation methodology itself.

Furthermore, Figure 10 implies a strong linear correlation between the filled out and freshly calculated concentrations, with a correlation coefficient of 0.990048, which is very close to 1, supporting this observation. It shows, that there was little error in the group individual calculations, as the filled out concentrations are very close to the freshly calculated concentrations.

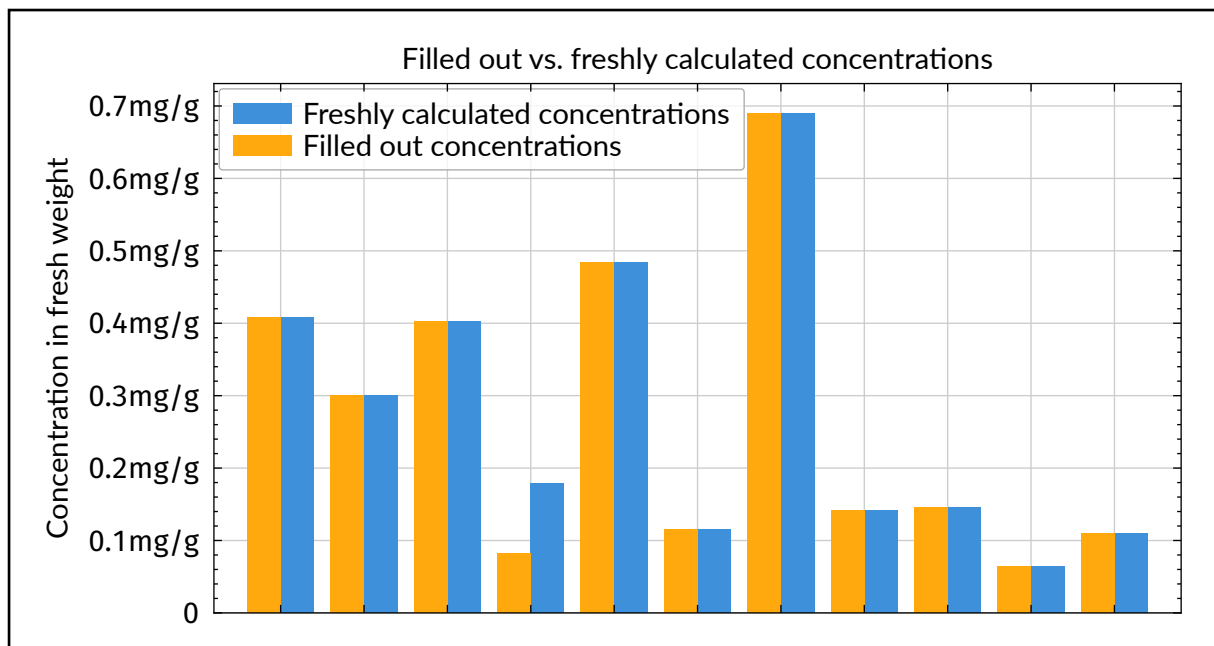


Figure 12: Comparison between filled out concentrations and freshly calculated concentrations

3.4.3.2 Direct 652nm formula & freshly calculated

The instructor's formulas in Equation 8 for calculating the total chlorophyll concentration describe, how to use the 652nm extinction coefficient for that purpose. Figure 10 shows the correlation between the freshly calculated concentrations and the concentrations calculated using the 652nm instructor's formula. There we can also see a strong linear correlation between the two sources, with a correlation coefficient of 0.987914.

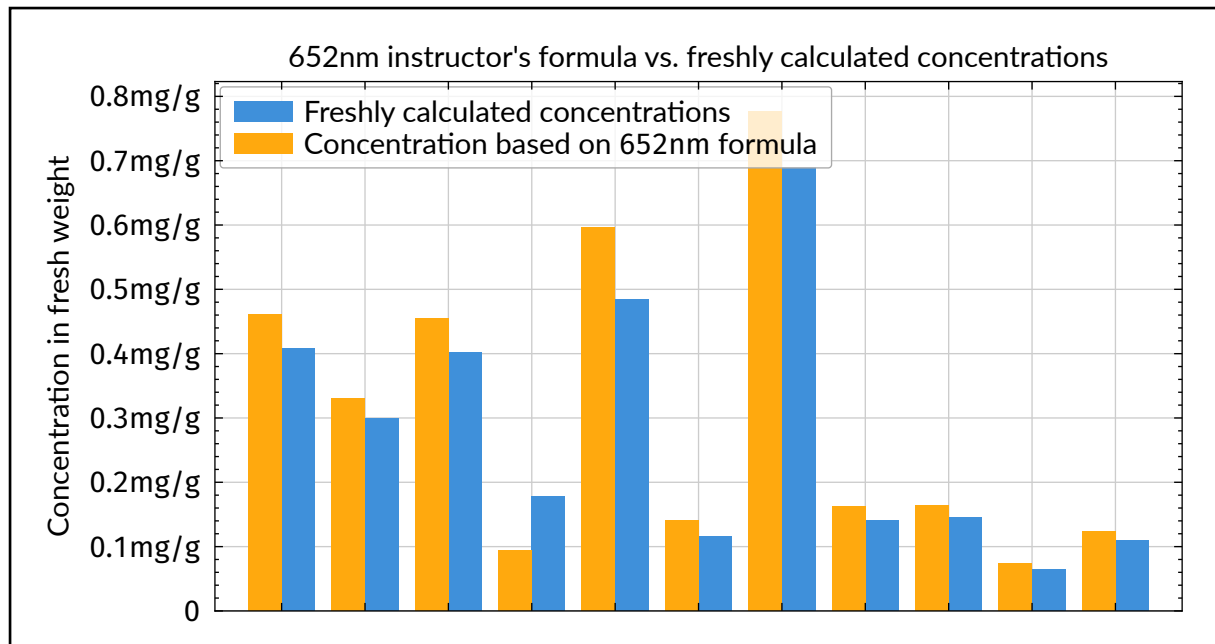


Figure 13: Comparison between 652nm instructor's formula and freshly calculated concentrations

Figure 13 shows that there is only a small difference between the two sources. Apart from the same outlier as in Section 3.4.3.1, there seems to be a positive offset in regards to the 652nm formula. This offset seems to grow proportionally to the concentration and therefore classifies it as a proportional offset.

Interestingly enough, at closer observation of Figure 10, a strong correlation between the 652nm instructor's formula and the filled out concentrations is visible. Indeed, the correlation coefficient is 0.997967, which is practically equivalent to 1.

All of the above observations lead us to conclude that the instructor's formula is very precise, but has a certain constant percentage accuracy offset. Slightly lowering the factor of 27.8 in the E_{652} formula might result in a more accurate calculation (assuming the addition of Chl_a and Chl_b is the accurate baseline).

Appendix

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